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JOURNAL OF  
ENVIRONMENTAL  
ECONOMICS AND  
MANAGEMENT

Journal of Environmental Economics and Management 46 (2003) 38–51

<http://www.elsevier.com/locate/jeem>

# Equity and efficiency in international markets for pollution permits

Leslie Shiell

*Department of Economics, University of Ottawa, 200 Wilbrod St., Ottawa, Ont., Canada K1N 6N5*

Received 15 May 2001; revised 4 February 2002

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## Abstract

The paper investigates the normative properties of international permit markets for uniformly mixed pollutants such as greenhouse gases. If permits can be traded at modest cost, there is unlikely to be any justification for second best arrangements that differentiate the marginal costs of abatement among countries. In particular, given a second best optimum, there will almost certainly be a cost-effective policy involving permit trading that dominates it in terms of social welfare, due to the redistributive effect of the initial allocation of permits as well as the efficiency gains of equalizing marginal costs. The socially optimal policy under permit trading may or may not be Pareto efficient, depending upon the specification of the model.

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*Keywords:* Tradable permits; Equity and efficiency; International pollution policy

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## 1. Introduction

The possibility of an international system of tradable permits to control greenhouse-gas emissions has stimulated interest in the normative properties of markets for such permits during the past decade. Much was already known, at least in theory, before the United Nations Framework Convention on Climate Change was signed in Rio in 1992. In particular, Montgomery [8] proved the cost-effectiveness of competitive permit markets for achieving a pre-specified abatement target. In contrast, relatively little attention was paid to the Pareto efficiency of permit markets before the 1990s, perhaps owing to the dominance of Coase's view that the distribution of property rights and efficient abatement were separable [4]. However, within the past decade, a literature has emerged investigating the Pareto efficiency of proposals for an international system of tradable permits to control greenhouse-gas emissions.

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*E-mail address:* [lshiell@uottawa.ca](mailto:lshiell@uottawa.ca).

The results of this literature can be summarized by the following four points, some of which are re-statements of standard results:

1. Pareto efficient abatement requires the equalization of marginal costs of abatement among countries [2,3].
2. In the second best (characterized here by the absence of lump-sum transfers), marginal costs of abatement are differentiated among countries [2,3,7].
3. For a given level of aggregate abatement, most initial allocations of permits will not lead to a Pareto efficient outcome [3].
4. The Pareto efficient level of abatement is dependent upon the distribution of permits [1,3].

The presentation of points 1 and 2 differs here from that of the cited authors. These authors refer to point 1 as Pareto efficiency with lump-sum transfers and point 2 as “constrained Pareto efficiency” or Pareto efficiency without lump-sum transfers. In contrast, the present paper opts for the more conventional distinction between Pareto efficiency and the second best. This usage highlights the consistency between these points and the theory of the second best, as developed by Lipsey and Lancaster [6].<sup>1</sup>

While the main focus of this new literature has been Pareto efficiency, a complete understanding of the performance of permit markets requires an integrated account of Pareto efficiency, the second best, and cost-effectiveness. The objective of the present paper is to provide such an account. In particular, the paper develops a diagrammatic presentation showing the relationships between the utility possibility frontiers and optimal policies under Pareto, second-best and cost-effective abatement in a simplified two-country model. It is shown that it will almost always be possible to find a cost-effective policy (equalized marginal abatement costs), associated with permit trading, that dominates a given second-best policy (differentiated marginal costs) in terms of social welfare, owing to the redistributive effect of the initial allocation of permits and the efficiency gains of equalizing marginal costs. Thus, assuming reasonable transactions costs for permit trading, there would appear to be no case for policies with differentiated pollution prices in efforts to curb global pollutants such as greenhouse gases.

The paper is divided into six Sections 2–7. Section 2 presents the notation and the model. Section 3 characterizes Pareto efficient abatement and relates point 4 to the efficiency conditions. Section 4 characterizes the utility possibility set under permit trading. Section 5 relates point 3 to cost-effectiveness, and compares the utility possibility frontiers for cost-effective, Pareto and permit-trading policies. Section 6 characterizes second-best abatement and compares the utility possibility frontiers for second best and Pareto efficient abatement. Finally, Section 7 compares the socially optimal policy under the second best with that under permit trading.

## 2. The model

The model is based on Chichilnisky and Heal [2]. There are  $N$  countries, denoted  $n = 1, \dots, N$ . Each country has a strictly concave utility function,  $u_n : \mathfrak{R}_+^2 \rightarrow \mathfrak{R}$ , which has as arguments the

<sup>1</sup>In a nutshell, the theory states that second-best optimality conditions deviate from the Paretian conditions (equalized marginal abatement costs in the present context).

private consumption good,  $c_n$ , and the state of the global atmosphere,  $\alpha$ . Production net of abatement costs in country  $n$  is represented as  $y_n \in [0, y_n^{\text{BAU}}]$ , where  $y_n^{\text{BAU}}$  is business-as-usual production—the maximum attainable level in the absence of pollution controls given technology and resources. Countries consume all of their production, net of abatement costs, transfers, and permit sales. There is no trade in the model.

Production causes emissions of greenhouse gases (or other global pollutants), denoted  $E_n$ , which diminish the quality of the atmosphere. Emissions in the absence of pollution controls are represented as  $E_n^{\text{BAU}}$ , which corresponds with business-as-usual production,  $y_n^{\text{BAU}}$ . Denote  $\alpha_n \in \mathfrak{R}_+$  as the abatement level undertaken by country  $n$ . Then

$$E_n = E_n^{\text{BAU}} - \alpha_n. \quad (1)$$

Atmospheric quality is equal to the aggregate abatement of all countries:

$$\alpha = \sum_{n=1}^N \alpha_n. \quad (2)$$

Abatement entails positive cost, which is represented as a reduction in  $y_n$ . In particular, the abatement function is

$$\alpha_n = \Phi_n(y_n), \quad (3)$$

with  $\Phi'_n < 0$  and  $\Phi''_n < 0$ . The marginal cost of abatement of country  $n$  ( $\text{MC}_n$ ) is equivalent to  $-\frac{dy_n}{d\alpha_n}$ .

Differentiating and rearranging (3) therefore yields

$$\text{MC}_n = -\frac{1}{\Phi'_n} > 0 \quad \text{and} \quad \frac{d\text{MC}_n}{d\alpha_n} = -\frac{d^2 y_n}{(d\alpha_n)^2} = \frac{\Phi''_n}{(\Phi'_n)^3} > 0, \quad (4)$$

which indicates that MC is positive and increasing in  $\alpha_n$ .

In an international permit regime, each country is granted an initial allocation of permits,  $\bar{E}_n$ , and then chooses its final level,  $E_n$ , in order to minimize the sum of the costs of abatement and permits. This choice entails equalizing  $\text{MC}_n$  and the price of permits, i.e.

$$-\frac{1}{\Phi'_n(y_n)} = P. \quad (5)$$

A country is a net purchaser or seller of permits depending upon whether its final emission level is greater or less than its initial allocation of permits. The individual budget constraint is therefore

$$c_n = y_n + P(\bar{E}_n - E_n), \quad (6)$$

where the second term on the right-hand side represents revenue from the sale of permits (positive for net sales, negative for net purchases). Market clearing requires balance between aggregate emissions and aggregate permits, i.e.  $\sum_n \bar{E}_n = \sum_n E_n$ .

If all countries face the same permit price, then marginal abatement costs are equalized in equilibrium, following (5), and the aggregate abatement cost is minimized. Thus, a uniform permit price results in a cost-effective arrangement of abatement activity, for any aggregate level of abatement (total number of permits).

### 3. Pareto efficient abatement

A Pareto efficient allocation is defined as a vector  $(c_1, y_1, \dots, c_N, y_N, \alpha)$  which solves the following problem:

$$\begin{aligned} \max_{\{c, y, \alpha\}} u_j(c_j, \alpha) \text{ subject to } & \text{(i) } u_n(c_n, \alpha) \geq \bar{u}_n, \quad \forall n \neq j, \\ & \text{(ii) } \alpha = \sum_{n=1}^N \Phi_n(y_n), \\ & \text{(iii) } \sum_{n=1}^N c_n = \sum_{n=1}^N y_n. \end{aligned}$$

Given the concavity assumptions, there is a unique solution to this problem for each combination of exogenous utility levels  $\bar{u}_n$ . Alternatively, the problem may be expressed in terms of welfare maximization:

$$\begin{aligned} \max_{\{c, y, \alpha\}} W = \sum_{n=1}^N \lambda_n u_n(c_n, \alpha) \text{ subject to } & \text{(i) } \alpha = \sum_{n=1}^N \Phi_n(y_n), \\ & \text{(ii) } \sum_{n=1}^N c_n = \sum_{n=1}^N y_n. \end{aligned}$$

In this formulation, the welfare weights are chosen by the social planner to sum to unity, i.e.  $\sum_n \lambda_n = 1$ . There is a unique solution to this problem for each combination of welfare weights, and one can trace out all the possible Pareto optima by varying the weights. The condition  $\sum_{n=1}^N c_n = \sum_{n=1}^N y_n$  entails that the planner has unrestricted access to lump-sum transfers to effect redistribution.

The second (welfare) formulation of the problem will be used here to derive the Pareto optimality conditions. Under the present model, the Lagrangian for this problem is

$$L = \sum_n \lambda_n u_n \left( c_n, \sum_n \Phi_n(y_n) \right) + \theta \left[ \sum_n y_n - \sum_n c_n \right].$$

The solution is characterized by the following two necessary conditions:

$$\lambda_n u_c^n = \theta \quad \forall n, \tag{7}$$

$$\Phi_j' = \frac{-\theta}{\sum_{n=1}^N \lambda_n u_\alpha^n} \quad \forall j. \tag{8}$$

Notice that from the definition of marginal cost (4), (8) can be rewritten as

$$MC_j = \frac{\sum_{n=1}^N \lambda_n u_\alpha^n}{\theta} \quad \forall j. \tag{8'}$$

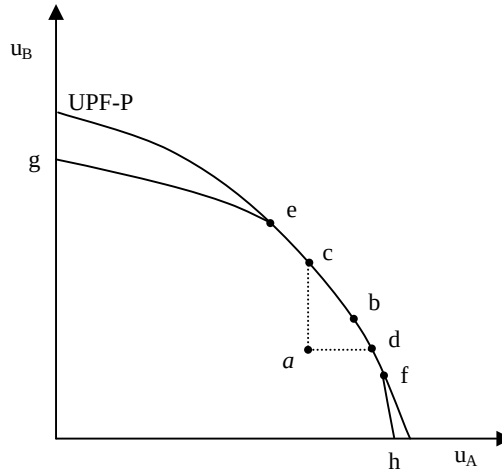


Fig. 1.

Then, substituting (7) into (8') and re-arranging yields the standard Lindahl–Bowen–Samuelson condition for the efficient production of a public good:

$$MC_j = \sum_{n=1}^N \frac{u_{\alpha}^n}{u_c^n} = \sum_{n=1}^N MRS_{\alpha,c}^n \quad \forall j. \quad (9)$$

where  $MRS_{\alpha,c}^n$  is the marginal rate of substitution of country  $n$  between environmental quality,  $\alpha$ , and private consumption,  $c$ . Since the right-hand side of (9) is independent of  $j$ , the marginal costs of abatement are equalized among all countries. This result reproduces Chichilnisky and Heal [2] and corresponds to point 1.

Point 4, concerning the dependence of the Pareto efficient level of abatement on the distribution of income or wealth, follows directly from the Lindahl–Bowen–Samuelson condition. Starting with an allocation that satisfies Eq. (9), a change in the distribution of income or wealth among countries will in general change the value of the right-hand side.<sup>2</sup> In order to restore equality, it will be necessary to alter marginal costs (the left-hand side) and therefore the aggregate level of abatement. This dependence simply reflects the fact that, for most countries, marginal willingness to pay for abatement varies with income.

Fig. 1 shows the utility possibility frontier with Pareto efficient abatement (UPF-P) in an economy consisting of two countries, noted A and B. The frontier has the standard shape, which follows from the assumption of concave utility. In the absence of regulation, the countries will be located below the frontier, at point  $a$  for example, owing to the public-good nature of abatement. To make the discussion interesting,  $a$  has been chosen so that country A is rich and country B is poor. Regulation to achieve Pareto-efficient abatement moves the countries onto the frontier. Different points on the frontier correspond with different distributions of income and therefore different aggregate levels of abatement.

<sup>2</sup> Under certain specifications of utility, changes in the MRS's of countries could be exactly offsetting under redistribution, despite differences in absolute levels of income, thus leaving the right-hand side of (9) unchanged. However, such specifications must be regarded as special cases.

By the definition of Pareto efficiency, movement along the frontier is accomplished by lump-sum redistribution. Nonetheless, there is one Pareto-efficient allocation associated with zero transfers, which is represented by point *b* in Fig. 1.<sup>3</sup> This point is located to the north-east of *a*, since both countries benefit from some level of pollution control, even given the existing imbalance in the distribution of income. Obviously, any point on the frontier between *c* and *d* represents a Pareto improvement over *a*, and therefore in theory one would expect negotiations over abatement to lead to a point on this arc (see [1]). In practice, it is probably unlikely that the poor country would be willing to make lump-sum payments to the rich; therefore, the relevant arc for negotiations is more likely *cb*.

#### 4. International permit trading

Unrestricted lump-sum transfers are a useful construct which scarcely exist outside the confines of economic theory. In recent years, aid between countries has usually been conditioned upon the fulfilment of certain performance criteria by the recipient, a situation referred to as tied aid.<sup>4</sup> To the extent that such tying of aid limits the extent of lump-sum transfers, it limits the possibility of movement along UPF-P. Further, in the context of an international permit system to control emissions of greenhouse gases, the question of aid to poor countries is subsumed into the question of the initial distribution of permits.

A permit-trading regime will be cost-effective, regardless of the level of aggregate abatement, provided permits trade at a uniform price. It is also possible to make the regime Pareto efficient, by choosing aggregate abatement and the distribution of permits simultaneously to satisfy the Lindahl–Bowen–Samuelson condition. In this case, the initial distribution of permits is equivalent to lump-sum transfers, as it represents transfers of wealth from net buyers to net sellers. However, the scope for such redistribution is restricted by the total number of permits, and therefore it will only be possible to reach a portion of UPF-P from the initial point *a* through the redistribution of permits. (In addition, each point on UPF-P represents a different total number of permits, due to the dependence of efficient abatement on distribution.)

The formal derivation of the utility possibility set under permit trading will make clear the relationship between UPF-P and permit trading. Substituting (1)–(3) and (6) into the utility function yields

$$u_n \left( y_n + P(\bar{E}_n - E_n^{\text{BAU}} + \Phi_n(y_n)), \sum_{n=1}^N \Phi_n(y_n) \right).$$

The equilibrium condition (5) defines each country's net output level as a function of *P*, i.e.  $y_n = y_n(P)$ . Therefore, utility levels are fully determined by *P* and the initial allocation of permits,  $\{\bar{E}_n\}_{n=1}^N$ , given levels of business-as-usual emissions,  $\{E_n^{\text{BAU}}\}_{n=1}^N$ . *P* is determined endogenously, as a result of equilibrium in the permit market, and therefore in practice the planner's choice variables are limited to the  $\bar{E}_n$ 's. However, mathematically the same result is obtained if *P* is treated as a choice variable subject to the constraint that the permit market clears. Thus, the

<sup>3</sup> This allocation will be characterized fully in Section 6.

<sup>4</sup> Consider, for example, the Highly Indebted Poor Countries Initiative of the World Bank and IMF.

utility possibility set can be obtained from the welfare maximization

$$\begin{aligned} \max_{\{P, \bar{E}\}} W &= \sum_{n=1}^N \lambda_n u_n \left( y_n + P(\bar{E}_n - E_n^{\text{BAU}} + \Phi_n(y_n)), \sum_{n=1}^N \Phi_n(y_n) \right) \quad \text{subject to} \\ \text{(i)} \quad &\sum_n \bar{E}_n = \sum_n (E_n^{\text{BAU}} - \Phi_n(y_n)), \\ \text{(ii)} \quad &\bar{E}_n \geq 0 \quad \forall n \end{aligned}$$

The non-negativity constraint on the initial allocation is straightforward. If negative allocations were permitted for some countries, then the system would be equivalent to unrestricted lump-sum transfers. It could be argued that the non-negativity constraint is not strong enough, in that it still allows allocations in which some countries receive revenues from permit sales in excess of their abatement costs. These excess revenues do not fit a strict definition of tied aid, as there is no corresponding performance criterion. Thus, it may be more realistic to restrict permit allocations such that no country receives more in net sales than its expenditure on abatement. Such a restriction would correspond to the concept of joint implementation, in which one country pays for abatement in another. On the other hand, excess revenues could be described as a reward for participating in the permit regime, in which case there would be a performance criterion. In any case, the qualitative results are similar. The simpler non-negativity constraint is used here for the sake of demonstration.

The Lagrangian for the problem is

$$\begin{aligned} L &= \sum_n \lambda_n u_n \left( y_n + P(\bar{E}_n - E_n^{\text{BAU}} + \Phi_n(y_n)), \sum_{n=1}^N \Phi_n(y_n) \right) \\ &+ \beta \left( \sum_n (E_n^{\text{BAU}} - \Phi_n(y_n)) - \sum_n \bar{E}_n \right). \end{aligned}$$

The solution is characterized by the first-order necessary conditions:

$$\begin{aligned} \frac{\partial L}{\partial P} &= \sum_n \lambda_n u'_c \left[ \frac{dy_n}{dP} + (\bar{E}_n - E_n^{\text{BAU}} + \Phi_n) + P \Phi'_n \frac{dy_n}{dP} \right] + \sum_n \lambda_n u'_z \left[ \sum_n \Phi'_n \frac{dy_n}{dP} \right] \\ &- \beta \sum_n \Phi'_n \frac{dy_n}{dP} = 0, \end{aligned} \quad (10)$$

$$\frac{\partial L}{\partial \bar{E}_n} = \lambda_n u'_c P - \beta \leq 0 \quad \text{and} \quad \bar{E}_n \geq 0 \quad \forall n, \quad \text{with complementary slackness.} \quad (11)$$

With two countries, A and B, there are three possible combinations of slack and binding constraints concerning the  $\bar{E}_n$ 's.

*Case 1:*  $\bar{E}_A, \bar{E}_B > 0$ .

In this case,  $\frac{\partial L}{\partial \bar{E}_n} = 0$  in (11), which yields

$$\lambda_n u'_c = \frac{\beta}{P}, \quad n = A, B. \quad (12)$$

Differentiating (5) and rearranging yields

$$\frac{dy_n}{dP} = \frac{1}{P^2} \frac{1}{\Phi_n''} < 0. \quad (13)$$

Then substituting (12) and (13) into (10) yields the Lindahl–Bowen–Samuelson condition:

$$\sum_{n=A,B} \frac{u_\alpha^n}{u_c^n} = P.$$

Therefore, the allocations in this case are Pareto efficient. From (12), the slope (absolute value) of the utility possibility frontier is derived as  $\frac{\lambda_A}{\lambda_B} = \frac{u_c^B}{u_c^A}$ .

These allocations correspond to the arc *ef* in Fig. 1. At point *b*, the initial allocation is such that each country uses all of its permits, without buying or selling. This point corresponds to the case of zero transfers in the previous section. It will also be shown, in Section 6, that this point corresponds to Negishi welfare weights. Then, as the welfare weights are varied in favour of country B,  $\bar{E}_B$  is increased and  $\bar{E}_A$  decreased, such that B becomes a net seller of permits—and therefore a recipient of transfers—and A becomes a buyer. This change is represented as a movement up UPF-P toward *e*. At the latter point, all permits are allocated to B, while  $\bar{E}_A = 0$ , which brings into play the non-negativity constraint. The opposite movement down UPF-P is symmetric in favour of A.

*Case 2:  $\bar{E}_A = 0$ ,  $\bar{E}_B > 0$ .*

In this case, all the permits are allocated to the poor country, B. From (11),  $\frac{\partial L}{\partial \bar{E}_A} < 0$  and  $\frac{\partial L}{\partial \bar{E}_B} = 0$ , which yields  $\lambda_A < \frac{\beta}{Pu_c^A}$  and  $\lambda_B = \frac{\beta}{Pu_c^B}$ . The slope of the utility possibility frontier is therefore  $\frac{\lambda_A}{\lambda_B} < \frac{u_c^B}{u_c^A}$ , which indicates that this portion of the frontier is below UPF-P. These allocations correspond to the arc *ge* in Fig. 1. Movement along *ge* is achieved by changing the total number of permits, i.e. aggregate abatement.<sup>5</sup>

*Case 3:  $\bar{E}_A > 0$ ,  $\bar{E}_B = 0$ .*

This case is symmetric to Case 2 and corresponds with the arc *fh* in Fig. 1.

To summarize, the utility possibility set under permit trading is bounded by the frontier *gef**h*, which will henceforth be referred to as the permit-trading frontier. The frontier has kink points at *e* and *f*. Above *e* (below *f*) all the permits are allocated to country B (country A).

<sup>5</sup> In order to get to the vertical axis, it may be necessary under some specifications to costlessly dispose of some of country A's output.

## 5. Cost-effective abatement

The observation by Chichilnisky et al. [3] that most initial allocations of permits will not lead to a Pareto efficient outcome, for a given level of aggregate abatement (point 3), is related to the criterion of cost-effectiveness. Point 1 establishes that cost-effectiveness (equalized marginal costs) is a necessary condition for Pareto efficiency. However, the reverse is not true: cost-effectiveness is not sufficient for Pareto efficiency. For that, the allocation must also satisfy (9), the Lindahl–Bowen–Samuelson condition.

Fig. 2 illustrates this relationship between cost-effectiveness and Pareto efficiency. Point  $i$  on the Pareto frontier corresponds to a particular choice of aggregate abatement and distribution of permits. It is then possible to redistribute these permits in a myriad of ways (infinitely many in fact). All of these arrangements will be cost-effective, provided permits trade at a uniform price, but only the original distribution, corresponding with point  $i$ , is Pareto efficient. The arc  $jk$ , which reproduces Chichilnisky et al. [3], indicates the possible combinations of utility that can be reached by this redistribution of permits and will henceforth be called the cost-effectiveness frontier conditioned on point  $i$ . This frontier is dominated at every point by UPF-P, with the exception of the tangency point  $i$ .

As drawn, the cost-effectiveness frontier passing through  $i$  does not cross the axes, which reflects the assumption that the total level of abatement is too small for the associated abatement costs to drive any country's utility to zero. In this case, the end points of the frontier correspond to allocations of all the permits to one country or the other, i.e. point  $j$  (point  $k$ ) corresponds to the allocation of all the permits to country B (country A). While actual properties will obviously depend upon the specifics of a given model, this assumption seems reasonable as a general rule when the aggregate abatement level is based upon a point on the Pareto frontier, for in this case one would not expect the aggregate abatement level to be very high, lump-sum transfers or permit trading being the preferred means of redistribution.

Of course, cost-effectiveness does not require that aggregate abatement be based upon a Pareto point. For example, the utility possibilities represented by the frontier  $lm$  reflect cost-effective

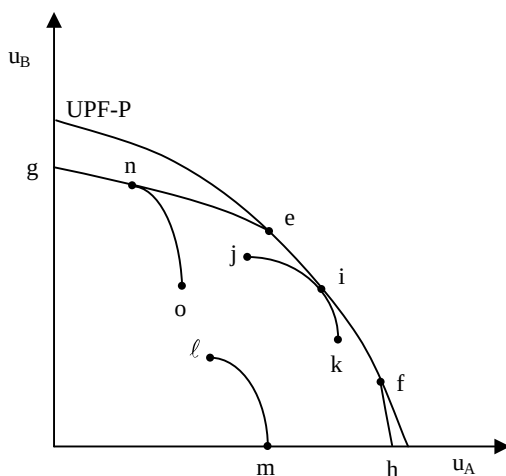


Fig. 2.

trading of permits for an inefficiently high level of abatement. As drawn, the frontier crosses the horizontal axis at  $m$ . At this point, most (or all) of the permits are allocated to the rich country, A. Coupled with the high level of abatement, this allocation implies that country B devotes all its income to abating emissions or buying permits, leaving nothing for consumption. Such a situation may be regarded as an extreme case of a bad policy implemented in a cost-effective manner. In general, the cost-effectiveness frontier may cross one or both of the axes for very high levels of abatement.

Clearly, there is a cost-effectiveness frontier corresponding to every point on UPF-P. However, only allocations within the permit-trading frontier  $gef$  can be reached with permit trading alone. Other allocations—above  $e$  on UPF-P for example—can only be reached with a combination of permit trading and lump-sum transfers. Thus, in the absence of lump-sum transfers, the permit-trading frontier  $gef$  is the envelope of all possible cost-effectiveness frontiers. The fact that all the permits are allocated to one country along the segments  $ge$  and  $fh$  indicates that cost-effectiveness frontiers built up against these segments but do not cross them. Such a case is illustrated by the frontier  $no$ .

## 6. Second-best abatement

In the absence of lump-sum transfers and permit trading, the planner seeks a second-best optimum. The utility possibility set is derived from the welfare maximization  $\max_{\{c, \alpha\}} W = \sum_{n=1}^N \lambda_n u_n(c_n, \alpha)$  subject to  $\alpha = \sum_{n=1}^N \Phi_n(c_n)$ . In this case, each country is limited to consuming what it produces, and therefore there is only one choice variable for each country. The solution to this problem yields the first-order conditions

$$\Phi'_j = \frac{-\lambda_j u'_c}{\sum_{n=1}^N \lambda_n u'_\alpha} \quad \forall j, \quad (14)$$

which, based on the definition of marginal cost (4), can be rewritten as

$$MC_j = \frac{\sum_{n=1}^N \lambda_n u'_\alpha}{\lambda_j u'_c} \quad \forall j. \quad (14')$$

Since there is no reason to expect the right-hand side of this expression to be constant across countries, marginal abatement costs are in general not equalized in the second best. This result reproduces Chichilnisky and Heal [2] and corresponds with point 2.

However, there does exist a special case in which marginal abatement costs are equalized in the second best. This case is related to point  $b$  in Fig. 1, the Pareto allocation for which transfers are optimally zero. This allocation corresponds with a particular choice of the welfare weights  $\{\lambda_n\}_{n=1}^N$ , and is characterized below. If a second-best policy is pursued using these weights, the absence of transfers does not bind, and one obtains the same result as when transfers are available.<sup>6</sup>

<sup>6</sup> It would appear that this is the allocation Martins and Sturm have in mind when they write, “typically there would be a Pareto point instead of a Pareto frontier” [7, p. 165].

However, for every other choice of welfare weights, the absence of lump-sum transfers in the second best does bind. Pollution control is the only remaining policy instrument, and therefore the planner responds by departing from Pareto efficiency in order to redistribute the benefits and costs of abatement among countries. In particular, the planner can affect the distribution of utility in two ways: through the allocation of the share of abatement among countries, which entails departing from equalized marginal abatement costs, and through the choice of total abatement.

The following proposition characterizes the Pareto efficient allocation with zero transfers.

**Proposition.** *The solution to the planner's welfare optimization without lump-sum transfers is Pareto efficient if and only if the welfare weights are Negishi weights.*

**Proof.** Negishi weights are defined as<sup>7</sup>

$$\lambda_j = \frac{1}{u_c^j} / \sum_{n=1}^N \frac{1}{u_c^n}. \quad (15)$$

Substituting this definition into the first-order condition for welfare maximization without lump-sum transfers (14') yields

$$MC_j = \sum_{n=1}^N \frac{u_{\alpha}^n}{u_c^n} \quad \forall j$$

which is identical to (9), the Lindahl–Bowen–Samuelson condition. Thus Negishi weights are sufficient for the attainment of Pareto efficient abatement without transfers.

Establishing necessity is only slightly more difficult. If the solution to the optimization without transfers is Pareto efficient, then (14') and (9) must be equivalent, i.e.

$$\frac{\sum_{n=1}^N \lambda_n u_{\alpha}^n}{\lambda_j u_c^j} = \sum_{n=1}^N \frac{u_{\alpha}^n}{u_c^n} \quad \forall j.$$

Re-arrangement yields

$$\lambda_j u_c^j = \frac{\sum_{n=1}^N \lambda_n u_{\alpha}^n}{\sum_{n=1}^N \frac{u_{\alpha}^n}{u_c^n}} \quad \forall j.$$

Now since the right-hand side of the expression above is independent of  $j$ , we have

$$\lambda_n u_c^n = \lambda_j u_c^j \quad \forall (n, j),$$

which yields

$$\frac{\lambda_n}{\lambda_j} = \frac{u_c^j}{u_c^n} \quad \forall (n, j). \quad (16)$$

Now by definition  $\sum_{n=1}^N \lambda_n = 1$ , or equivalently  $\sum_{n=1}^N \frac{\lambda_n}{\lambda_j} = \frac{1}{\lambda_j}$ . Substituting (16) into the latter expression and re-arranging yields the definition of the Negishi weight, (15).

<sup>7</sup>The original reference is [9]. See also [5] for an accessible introduction.

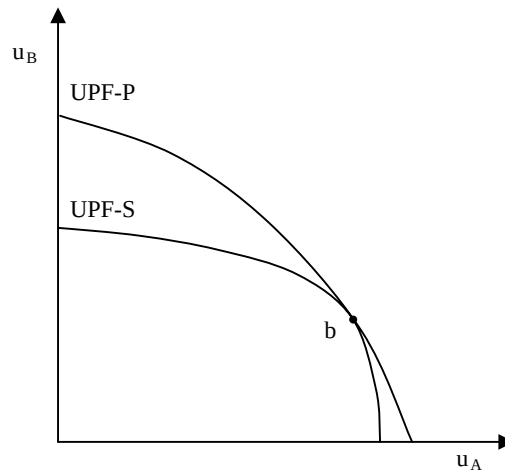


Fig. 3.

Fig. 3 compares the utility possibility frontier in the second best (UPF-S) with that under Pareto efficiency (UPF-P). The two frontiers are tangent at  $b$ , which corresponds to the case of Negishi welfare weights. Everywhere else, UPF-S lies within UPF-P, due to the inefficiency of the second-best policy. Put another way, for any allocation on UPF-S other than  $b$ , it is always possible to achieve a Pareto improvement through the introduction of lump-sum transfers and the rearrangement of abatement policy.

## 7. Welfare optimization

Fig. 4 brings together the Pareto, second-best and permit-trading frontiers. The standard argument in favour of the second best is that the absence of lump-sum transfers means that second-best policies are the only means to redistribute utility and thus attain equity goals. This argument is blunted in the context of international permit trading, as demonstrated by a comparison of the second best and the permit-trading frontiers. Fig. 4 shows social welfare indifference curves corresponding to optimal policy under the second best,  $W_s$ , and international permit trading,  $W_p$ .<sup>8</sup> These welfare levels may also be compared with the Pareto efficient allocation without transfers or trading,  $b$ .

Point  $b$  is not very attractive in terms of global equity. Therefore if transfers or trading were not possible, the planner would prefer the second best solution at  $c$ , where the welfare indifference curve  $W_s$  is tangent to UPF-S. This solution involves increasing abatement in the rich country and decreasing it in the poor country, so that  $MC^A > MC^B$ . Aggregate abatement may also be different at  $c$  than  $b$ . This solution is not cost effective, but in the absence of feasible transfers or permit trading it would be the only way to achieve a more equitable distribution of utility. In other words, the classic trade-off between equity and efficiency would hold.

<sup>8</sup> These indifference curves do not necessarily correspond to the welfare function used in previous sections, since the latter is just an expedient for tracing out the utility possibility sets.

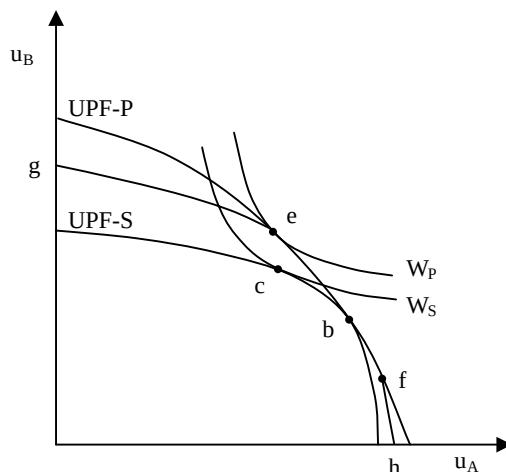


Fig. 4.

Under the system of tradable permits, the optimal allocation occurs at point  $e$ , which represents a higher level of welfare,  $W_P$ . This point is Pareto efficient and, as drawn, occurs at a kink in the frontier. Depending upon the sharpness of the kink, this point will be optimal for a variety of specifications of welfare and therefore has the potential to reconcile different ethical views concerning the relative weighting of rich and poor countries in global pollution policy. This point would seem to correspond to the outcome intended by the authors of the Clean Development Mechanism of the Kyoto Protocol. Of course, there is no guarantee that the permit optimum will be located at the kink. For a steeper social indifference curve it would be located down  $ef$ , while for a flatter social indifference curve it would be located on  $ge$ . In the latter case, the policy would not be Pareto efficient but it would be cost-effective.

## 8. Conclusion

The paper has shown how the properties of international permit markets relate to the three normative criteria Pareto efficiency, second best, and cost-effectiveness. In terms of policy, this treatment suggests that there is probably no role for second best arrangements that differentiate the marginal costs of abatement among countries, assuming the transactions costs of permit trading are modest. In particular, given a second best optimum, there will almost certainly be a cost-effective policy involving permit trading that dominates it in terms of social welfare. This result follows from the lump-sum redistributive nature of the initial allocation of permits as well as the efficiency gains of equalizing marginal costs. Thus, it would appear that the traditional trade-off between equity and efficiency is absent, at least in theory, in international permit systems to control globally mixed pollutants such as greenhouse gases.

This result depends critically upon the rather heroic assumption that there are no other sources of inefficiency operating simultaneously, such as market power, missing markets, etc. In the real world, of course, such problems do exist. If one wishes to take them into account while modeling pollution policy, then it will be necessary to consider the number and nature of the instruments

available to the policy maker. If, as experience would suggest, the number of instruments is quite limited, then one is back in the realm of the second best, in which very few systematic results can be derived from theory.

Finally, it is worth emphasizing the limitations of the traditional criteria in making recommendations for international pollution control. The derivation of optimal policy, whether Pareto efficient or second best, depends upon the ability to estimate reliable measures of marginal willingness to pay for a cleaner environment. Since our ability in this regard remains limited, it would appear that the only practical guide for policy at present is cost-effectiveness.

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